

Smart Livestock,

Greener Future

MooMe is an IoT-AI platform that monitors livestock health, automates farm operations, and reduces greenhouse gas emissions from dairy farming, contributing to Rwanda's NDC goals and FAO climate targets.

\$723,287	\$205,479	\$246,575	\$271,233
TOTAL BUDGET (3 YEARS)	YEAR 1	YEAR 2	YEAR 3

YEAR 1: PROTOTYPE STARTUP	YEAR 2: PROOF OF CONCEPT	YEAR 3: GROWTH TO SCALE	GRAND TOTAL
\$205,479	\$246,575	\$271,233	\$723,287

Reducing Greenhouse Gas Emissions from Livestock through IoT

According to FAO, livestock contributes 14.5% of global GHG emissions. MooMe's IoT sensors detect harmful gases produced from manure and urine, enabling early mitigation and supporting Rwanda's Nationally Determined Contributions (NDC) and green project goals.

GHGR

Greenhouse Gas Reduction through IoT-enabled detection and biogas mitigation

FAO

Aligned with FAO goals to reduce livestock sector's contribution to global warming

— PROJECT OBJECTIVES

Start Small, Think & Grow Big

A comprehensive IoT-AI platform covering environmental sustainability, animal health, productivity, and smart farm automation.

01

GHG Emissions Reduction

Detect CH₄, NO₃⁺, CO₂, NH₃ from manure and urine. Deploy biogas technology with IoT-monitored pressure and temperature sensors as mitigation.

02

Heat Stress Management

Monitor cow body temperature (IR sensors), outside temperature, humidity, and ventilation. Auto-activate cooling fans and water spray systems.

03

Water Efficiency

Per-cow water flow sensors track intake. Smart pumps ensure optimal hydration while minimizing waste through AI-driven scheduling.

04

Feed Utilization & Intake

Load cells measure feed consumption per cow. AI optimizes nutrition based on age, weight, lactation stage, and production goals.

05

Disease Detection & Health

AI analyzes temperature patterns, activity levels, rumination rate, and resting time to predict diseases early, before visible symptoms appear.

06

Reproductive System Monitoring

Track heat cycles, activity spikes, and behavioral changes. AI alerts farmers to optimal breeding windows, improving conception rates.

07

Rumination & Resting Analysis

Accelerometer-based activity sensors measure rumination rate and resting time, which are key indicators of digestive health and overall well-being.

08

Milk Production Optimization

Smart milk meters per cow track yield. AI predicts production trends, recommends optimal milking times, and detects quality issues.

Smart Ventilation Control

Automated fan speed adjustment based on real-time temperature, humidity, and air quality data to maintain optimal barn conditions 24/7.

Real-Time Monitoring at Every Level



Body Temperature

Non-contact IR sensors per cow for continuous fever and health monitoring



Water Intake

Flow sensors per cow track daily consumption and detect dehydration



Feed Measurement

Load cells + HX711 modules measure individual feed intake precisely



Milk Metering

Smart meters record per-cow yield with AI-optimized milking schedules



Gas Detection

MQ-135 + O₂ sensors detect CH₄, NH₃, CO₂ from waste and environment



Environment

DHT22 sensors for temperature, humidity, and air quality monitoring



RFID Identification

Unique RFID tags per cow for individual tracking and data linkage

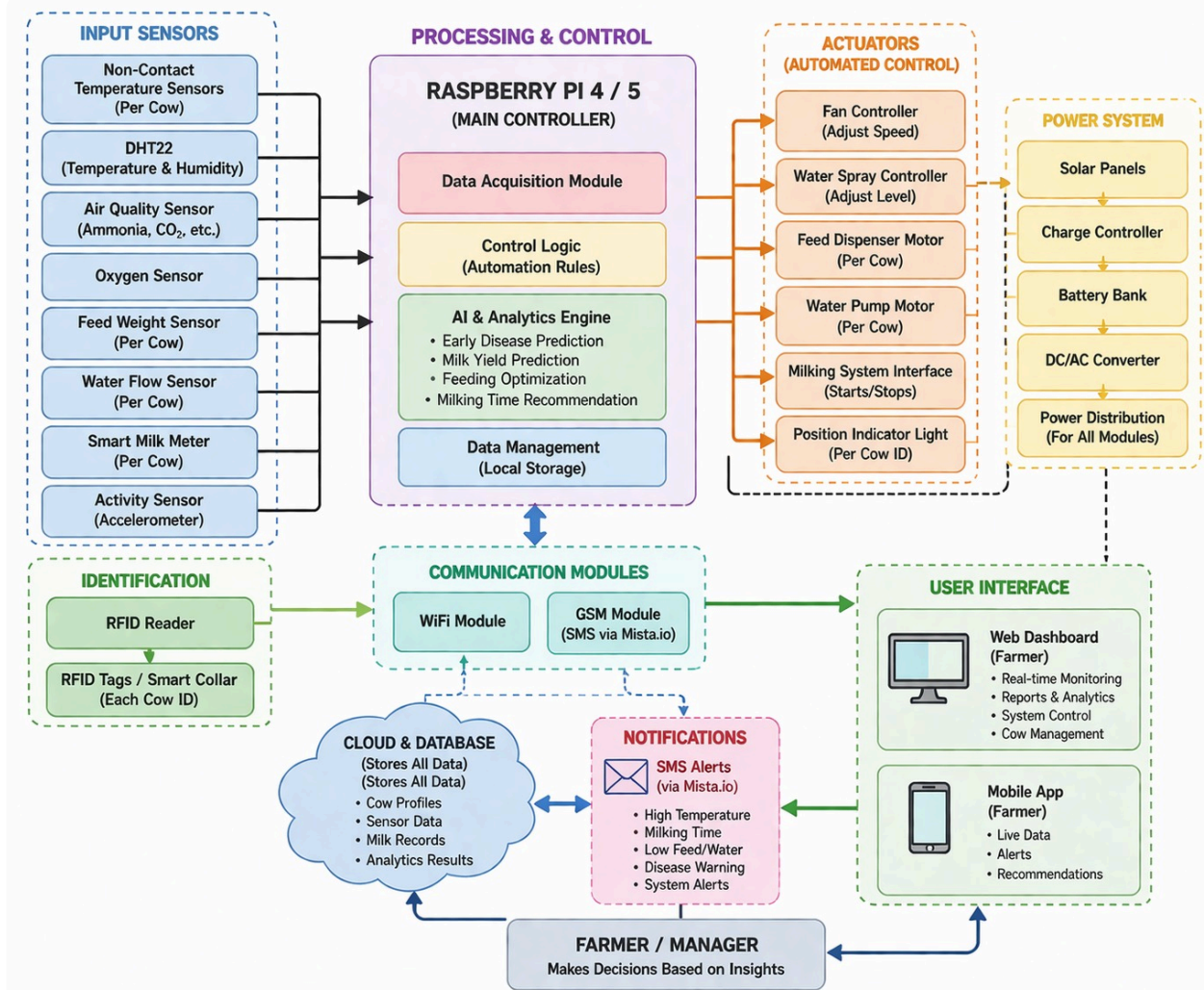


Solar Power

200W solar panel + battery backup for 24/7 off-grid operation

End-to-End IoT-AI Platform

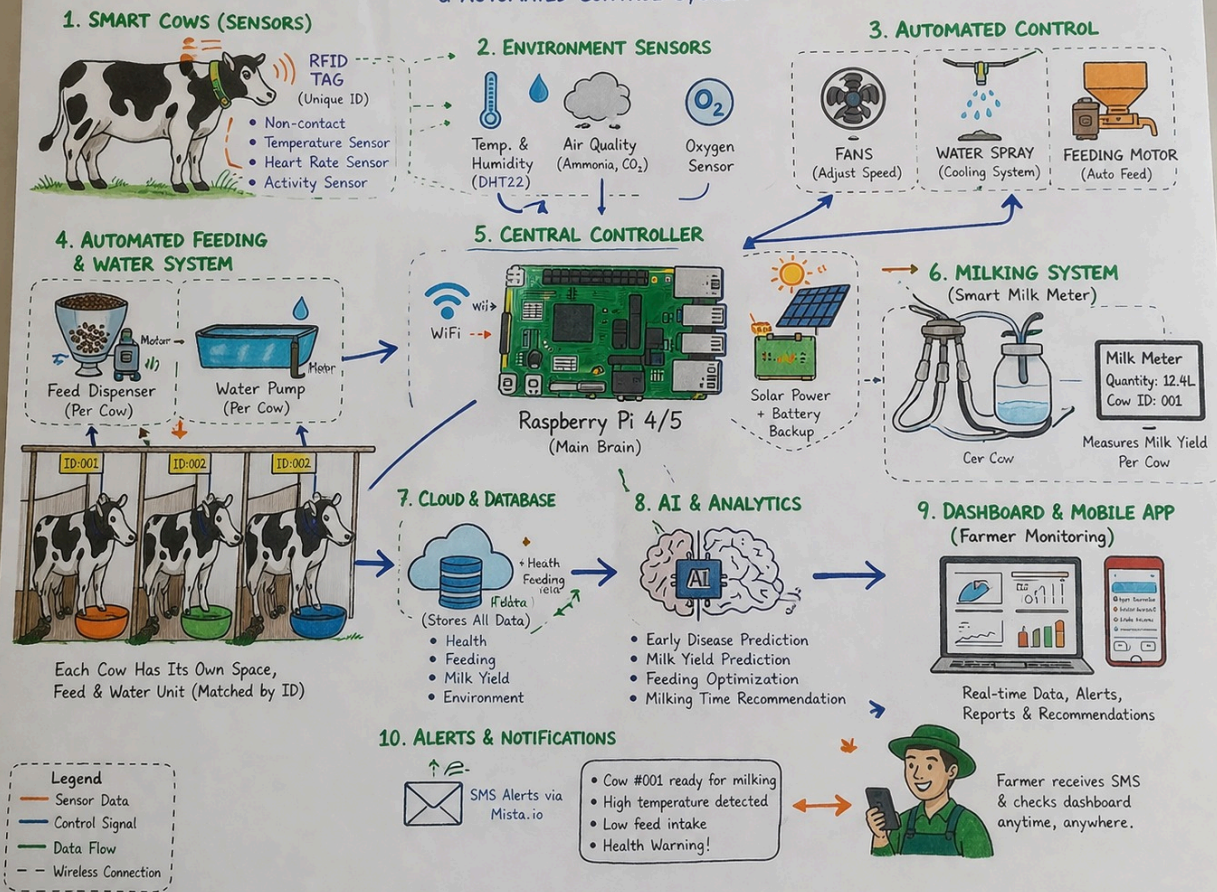
From sensor data collection to AI-driven insights and automated control. A complete smart farming ecosystem.



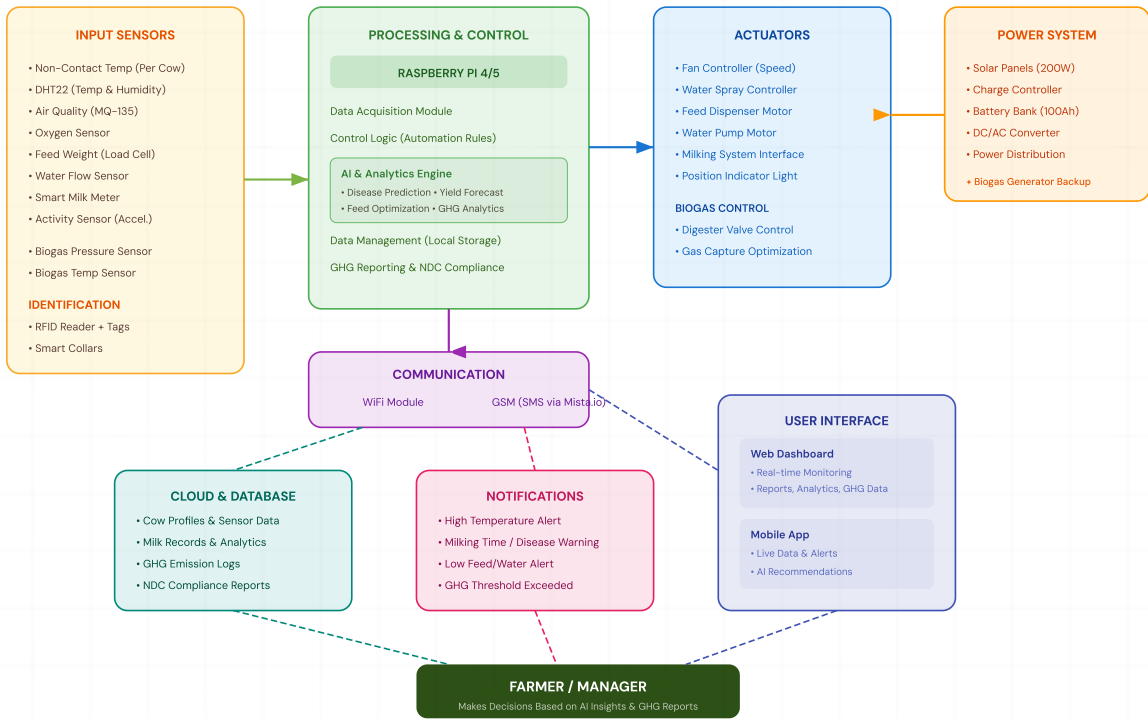
SYSTEM BLOCK DIAGRAM

MooMe^{Wi-Fi}

IoT - AI LIVESTOCK MONITORING & AUTOMATED CONTROL SYSTEM



IOT-AI MONITORING WORKFLOW



Quarterly Budget Breakdown

Solution Design, Development & Promotion: \$723,287 USD across 3 years, split into quarterly allocations per year.

CATEGORY	ACTIVITY / DELIVERABLE	YEAR 1 \$205,479				Y1	YEAR 2 \$246,575			
		Q1	Q2	Q3	Q4		Q1	Q2	Q3	Q4
SOLUTION DESIGN	System Architecture & Requirements	\$8,000	\$5,000	—	—	\$13,000	\$4,000	\$2,000	—	
	IoT Sensor Design & GHG Sensor Spec	\$6,000	\$5,000	\$2,000	—	\$13,000	\$3,000	\$3,000	\$2,000	
	AI/ML Model Design (Disease, GHG, Yield)	\$5,000	\$5,000	\$3,000	\$2,000	\$15,000	\$4,000	\$4,000	\$3,000	\$2,000
	GHG / Biogas Monitoring System Design	\$4,000	\$4,000	\$2,000	—	\$10,000	\$3,000	\$3,000	\$2,000	\$2,000
	Dashboard & Mobile App UX/UI Design	\$3,000	\$3,000	\$2,000	\$1,000	\$9,000	\$2,000	\$2,000	\$2,000	\$1,000
	Design Subtotal	\$26,000	\$22,000	\$9,000	\$3,000	\$60,000	\$16,000	\$14,000	\$9,000	\$5,000
SOLUTION DEVELOPMENT	IoT Hardware Procurement & Assembly	\$3,000	\$10,000	\$8,000	\$4,000	\$25,000	\$6,000	\$8,000	\$7,000	\$5,000
	Embedded Software & Firmware Dev	\$2,000	\$4,000	\$5,000	\$4,000	\$15,000	\$4,000	\$5,000	\$5,000	\$4,000
	AI Model Training & Deployment	—	\$3,000	\$5,000	\$5,000	\$13,000	\$5,000	\$6,000	\$6,000	\$5,000
	Web Dashboard & Mobile App Dev	—	\$3,000	\$5,000	\$5,000	\$13,000	\$5,000	\$6,000	\$5,000	\$5,000
	GHG Monitoring, Biogas & NDC Integration	\$1,000	\$3,000	\$4,000	\$4,000	\$12,000	\$4,000	\$5,000	\$5,000	\$4,000
	Cloud Infrastructure, Database & APIs	\$2,000	\$3,000	\$3,000	\$4,479	\$12,479	\$4,000	\$4,000	\$4,000	\$4,579
	Testing, QA & Pilot Deployment	—	—	\$5,000	\$7,000	\$12,000	\$4,000	\$5,000	\$5,000	\$5,000
	Development Subtotal	\$8,000	\$26,000	\$35,000	\$33,479	\$102,479	\$32,000	\$39,000	\$37,000	\$32,579
SOLUTION PROMOTION	Farmer Training & Capacity Building	—	—	\$3,000	\$4,000	\$7,000	\$3,000	\$3,000	\$4,000	\$3,000
	Stakeholder Engagement & Live Demos	\$1,000	\$2,000	\$2,000	\$2,000	\$7,000	\$2,000	\$3,000	\$3,000	\$3,000
	NDC / GHG Impact Reports & Publications	—	—	\$2,000	\$3,000	\$5,000	\$2,000	\$2,000	\$3,000	\$3,000
	Marketing, Branding & Conferences	\$2,000	\$2,000	\$3,000	\$3,000	\$10,000	\$3,000	\$3,000	\$3,000	\$3,000
	Partnership Dev & Green Fund Applications	\$3,000	\$3,000	\$4,000	\$4,000	\$14,000	\$4,000	\$4,000	\$4,000	\$4,000
	Promotion Subtotal	\$6,000	\$7,000	\$14,000	\$16,000	\$43,000	\$14,000	\$15,000	\$17,000	\$16,000
QUARTERLY TOTALS		\$40,000	\$55,000	\$58,000	\$52,479	\$205,479	\$62,000	\$68,000	\$63,000	\$53,579

IoT Kit: 12 Cow System

Hardware component breakdown for the prototype startup phase. Prices in USD
(2,699,000 RWF total, included in the \$723,287 global budget).

~\$2,300

2,699,000 RWF per 12-cow unit

#	COMPONENT	QTY	UNIT (USD)	TOTAL (USD)
1	Raspberry Pi 4/5	1	\$102	\$102
2	MicroSD Card (64GB)	1	\$13	\$13
3	Power Supply	1	\$9	\$9
4	Non-contact Temp Sensor (IR)	12	\$17	\$204
5	DHT22 Sensor	2	\$7	\$14
6	Air Quality Sensor (MQ-135)	1	\$13	\$13
7	Oxygen Sensor	1	\$21	\$21
8	RFID Reader Module	1	\$15	\$15
9	RFID Tags	12	\$2	\$20
10	Water Flow Sensor	12	\$9	\$102
11	Load Cell + HX711	12	\$13	\$153
12	Smart Milk Meter	12	\$51	\$611
13	DC Water Pump	12	\$17	\$204
14	Fans (Cooling)	4	\$21	\$85
15	Relay Modules	5	\$7	\$34
16	Motor (Feed Dispenser)	12	\$15	\$184
17	Pipes & Spray Nozzles	1 set	\$51	\$51
18	GSM Module (SIM800L)	1	\$13	\$13
19	Wi-Fi Router	1	\$30	\$30
20	Solar Panel (200W)	1	\$128	\$128
21	Battery (100Ah)	1	\$153	\$153
22	Charge Controller	1	\$34	\$34
23	Inverter	1	\$68	\$68
24	Wires & Installation	1	\$39	\$39

Total Prototype Hardware (12 Cows)

\$2,300

AI-IoT Digital Solutions Contributing to NDC Program & GHG Reduction

Livestock GHG Detection

Real-time monitoring of CH₄, NO₃⁺, CO₂, and NH₃ emissions from manure, urine, and cow dung using IoT gas sensors, enabling data-driven mitigation before pollution occurs.

Biogas Technology

Converting livestock waste gases into clean biogas energy. IoT sensors monitor digester pressure and temperature to optimize methane capture and reduce atmospheric emissions.

NDC Compliance

Automated reporting aligned with Rwanda's Nationally Determined Contributions under the Paris Agreement. Verifiable GHG reduction data for green financing and carbon credits.

FAO Alignment

Directly contributes to FAO's goal of reducing the livestock sector's 14.5% contribution to global GHG emissions through technology-driven early detection and waste management.

Green Project Financing

IoT-verified emission reduction data qualifies MooMe farms for green climate funds, carbon credit markets, and ESG-aligned investment, creating financial incentives for sustainable farming.

Circular Economy

Waste-to-energy cycle: livestock waste produces biogas for farm energy, reducing both GHG emissions and energy costs. Digestate becomes organic fertilizer, completing the loop.

Start Small, Think & Grow Big



MooMe

IoT-AI Livestock Monitoring Platform. Smart farming for a greener, more productive future. Contributing to Rwanda's NDC goals and FAO global warming targets through technology-driven emission reduction.

Component 4: Technical Support / IoT • PI/E • Total Budget: \$723,287 USD (3 Years)

PROTOTYPE STARTUP • 2026